

Reversible β -Sheet-like Ordering and XPCS-Resolved Dynamic Arrest during Elastin-like Polypeptide Phase Separation

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Abstract

Amyloid-like β -sheet order is usually associated with persistent aggregates, yet related contacts can also act as reversible physical junctions. Temperature-dependent SAXS/WAXS of sequence-defined (VPAVG)₃₀ reveals a reversible transition from a monodisperse state at 10 °C to a hierarchical assembly at 30 °C, with an approximate $I(q) \propto q^{-3.2}$ low- q response and peaks at $q = 1.38$ and $0.72 \text{ \AA}^{-1}$, compatible with β -sheet-like packing. In a separate isothermal experiment on a freshly loaded sample,

SA-XPCS at 30 °C shows a two-step correlation decay whose fast-decay amplitude falls progressively across 76–167 nm over 47 min, indicating an increasing fraction of the scattering ensemble is arrested within the 1.6 s maximum accessible delay. Thermally reversible β -sheet-like local order therefore coexists with progressive mesoscale arrest, indicating that amyloid-like packing need not imply irreversible aggregation and offering a route to thermally switchable protein networks.

Amyloid-like β -sheet ordering occupies an unusual position in protein science and polymer materials. Pathological amyloid fibrils formed by unrelated proteins and peptides are associated with Alzheimer’s disease,¹ Parkinson’s disease,² type II diabetes,³ prion diseases,⁴ and systemic light-chain amyloidosis.⁵ Many amyloid fibrils are kinetically persistent because extended hydrogen-bonded sheets are reinforced by tightly packed steric-zipper interfaces.⁶ Yet amyloid-like order is not intrinsically pathological: functional amyloids store mammalian peptide hormones in pituitary secretory granules⁷ and build the extracellular matrix of bacterial biofilms,⁸ and related assemblies are increasingly used as mechanically robust biomaterials.⁹ What differs across these systems is not whether β -sheet-like order forms but how long it lasts: hormone granules disassemble to release functional monomer, whereas steric-zipper cores are far more persistent. Amyloid-like packing therefore spans a wide range of stabilities rather than committing a sequence to one permanent structure,¹⁰ and where a sequence falls in that range is a materials-design variable.

Reversible β -sheet-like contacts are attractive as candidate physical junctions because backbone hydrogen bonding can stabilize locally ordered assemblies, whereas hydrated or weakly packed interfaces can remain dynamic or reversible.^{10–13} Elastin-like polypeptides (ELPs) are well suited to test this possibility because their sequence, molecular weight, and lower critical solution temperature (LCST) behavior can be programmed recombinantly.^{14–18} Above the LCST, ELPs undergo hydrophobic desolvation and can form coacervates or hierarchical assemblies depending on sequence, concentration, and thermal history.^{17–19} In particular, alanine-containing ELPs can form stiff, thermoresponsive gels through arrested

phase separation.^{20,21} Building on that work, the complementary question addressed here is whether local β -sheet-like packing can emerge reversibly while the mesoscale assembly progressively loses mobility.

Resolving whether a structurally assembled state remains dynamically evolving requires a dynamical measurement. In colloidal gels, microscopic relaxation can continue to evolve after the time-averaged structure changes only weakly, a nonequilibrium process commonly termed physical aging.^{22,23} Related heterogeneous relaxation has been resolved during protein liquid-liquid phase separation,²⁴ protein gelation,²⁵ and cage relaxation in crowded protein solutions.²⁶ During arrested phase separation, a protein-rich morphology can therefore continue to reorganize through junction exchange, stress relaxation, and residual coarsening even when its SAXS profile changes modestly. SA-XPCS probes that evolution directly through speckle decorrelation at the same low- q length scales sampled by SAXS.

Here, temperature-dependent SAXS/WAXS and isothermal SA-XPCS provide complementary structural and dynamical views of concentrated aqueous (VPAVG)₃₀. At 10 °C the scattering is dominated by molecular dimensions with no detectable mesoscale assembly; heating to 30 °C produces hierarchical low- q scattering and high- q peaks compatible with β -sheet-like packing, and cooling restores the low-temperature profile. In a separate experiment on a freshly loaded sample, isothermal SA-XPCS at 30 °C reveals a two-step decay whose fast fraction falls progressively with elapsed time, so the central result is a reversible structural transition coexisting with a time-dependent loss of mesoscale mobility.

Figure 1 summarizes the SA-XPCS experiment. Its upper-left cartoon illustrates one candidate pathway, not a direct observation: protein-poor solvent regions are hypothesized to appear and grow within the slower protein-rich phase, displacing and compressing it toward a coarsened, network-like morphology. This dynamic-asymmetry picture follows Tanaka’s viscoelastic phase-separation framework and is consistent with prior scattering studies of arrested ELP gels.^{20,27} The (VPAVG)₃₀ sample (approximately 100 mg mL⁻¹) contained 20 mmol L⁻¹ HEPES (pH 8), 250 mmol L⁻¹ NaCl, and 2 mmol L⁻¹ DTT. Each 2 s

SA-XPCS acquisition at APS 8-ID-I (Experimental Section) used a previously unexposed sample position; temporal and azimuthal averaging yielded $I(q)$ and multi-tau correlation analysis yielded $g_2(q, \Delta t)$.

Figure 2 establishes the low- and high-temperature structures and their reversibility. Two aliquots from the same stock were sealed in 2 mm quartz capillaries and measured at APS 12-ID-B using simultaneous SAXS/WAXS detection at 13.3 keV.^{28–30} The aliquot labeled “Measurement” had undergone more than 10 heating/cooling cycles, whereas the “Reference” aliquot had remained at 4 °C. At 10 °C, the two profiles overlap. A Guinier fit to the Reference data gives an apparent $R_g = 22.65 \pm 0.50$ Å (Figure S7), consistent with a dispersed, unassembled low-temperature state lacking detectable mesoscale aggregates; this profile is recovered after repeated cycling. The 10 °C profiles span $0.0081 < q < 1.41$ Å^{−1} and the 30 °C profiles $0.0046 < q < 1.61$ Å^{−1}, so the absence of mesoscale assembly at 10 °C is established for structures below approximately 78 nm. At 30 °C, both aliquots reproducibly develop an approximate $I(q) \propto q^{-3.2}$ response at low q (slopes fitted over $0.012 < q < 0.040$ Å^{−1}: −3.17 for the Reference and −3.15 for the Measurement aliquot) and the same pair of high- q features, at $q = 1.38$ and 0.72 Å^{−1} in the Reference aliquot and 1.36 and 0.74 Å^{−1} in the Measurement aliquot. Their close overlap at both temperatures shows reproducible interconversion within the sensitivity of the scattering measurements. A more direct test at 8-ID-I took one aliquot—rather than two compared to each other—through seven consecutive 6–34 °C cycles at 1 °C min^{−1} (Figure S6). Across the seven cycles the 6 °C profile recovers to within a 9% relative standard deviation and the 33.8 °C profile to within 5%, with no systematic drift, so the transition is repeatable on one and the same material. The spread at the mid-ramp 31.9 °C state is larger, 33%, but that is a sampling artifact of crossing the steep part of the transition at a slightly different temperature in each cycle, and falls to 7% once that is accounted for (SI Section 6.2). Video S1 independently shows recovery of optical transparency after cooling.

The two reciprocal-space regimes describe a hierarchical assembly. Power-law small-angle

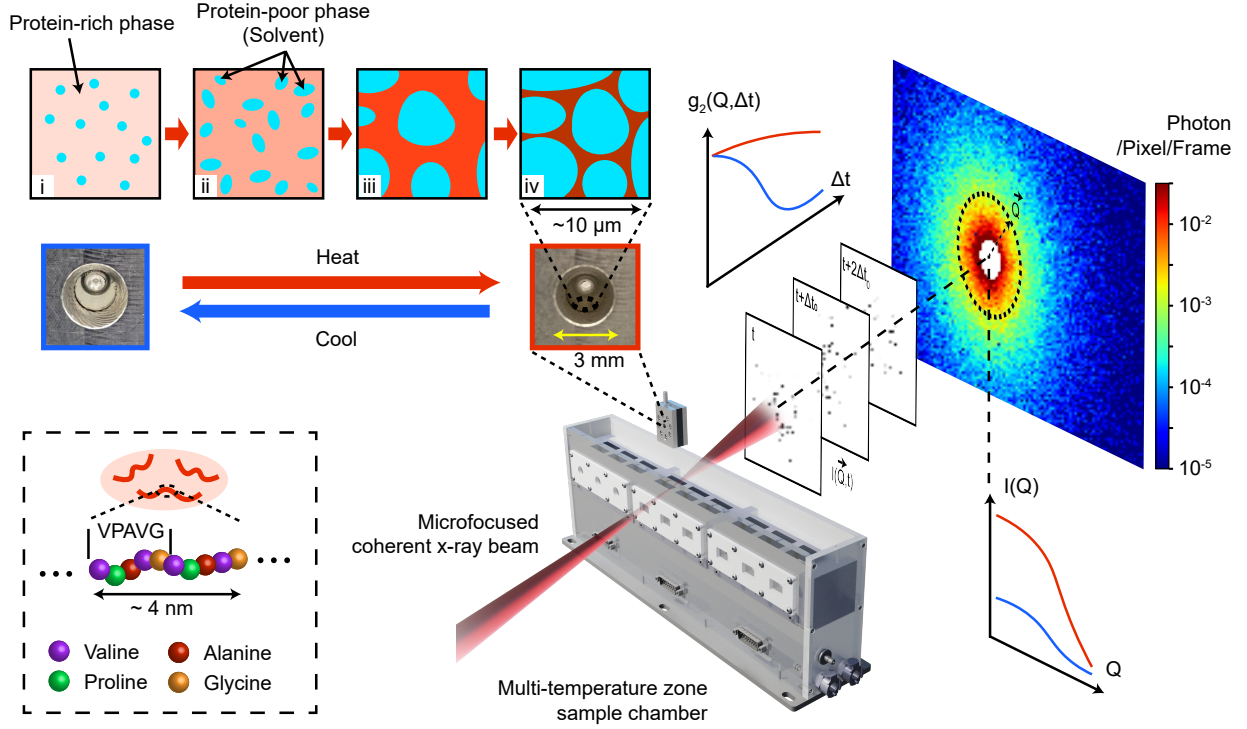


Figure 1: SA-XPCS measurement of thermally induced (VPAVG)₃₀ assembly at APS 8-ID-I. A microfocused coherent x-ray beam probes the (VPAVG)₃₀ solution (lower left: one repeat unit, residues colored as keyed, with its extended length) held in an aluminum cell (center; its liquid chamber is 3 mm in diameter, shown below and above the transition and linked by the heating/cooling arrows) that seats in direct conductive contact with a Peltier-controlled temperature zone of the multi-temperature-zone sample chamber. The detector records a sequence of 20 μ s frames, each registering only sporadic photon counts, and the same frames feed two analyses: averaging them yields the SAXS intensity $I(q)$, which reports the mesoscale and molecular structure formed on heating, and correlating them in time yields $g_2(q, \Delta t)$, which reports the relaxation of low- q mesoscale density fluctuations. Both are sketched schematically beside the detector (red: assembled state; blue: dispersed state; curves are illustrative, not data). Right: the frame-averaged detector image, color bar in photons per pixel per frame, white center the beamstop mask; its graininess reflects the sparse photon statistics rather than resolved speckle. Upper left, stages i–iv (the 10 μ m bar gives the beam footprint for scale): a conceptual illustration, not a real-space observation, of the candidate viscoelastic phase-separation pathway motivated by Tanaka’s framework and prior scattering studies of arrested ELP gels,^{20,27} in which protein-poor solvent pools (cyan) are hypothesized to nucleate and grow within the still-continuous protein-rich phase (deepening red denotes increasing protein concentration), driving it toward a coarsened, network-like morphology.

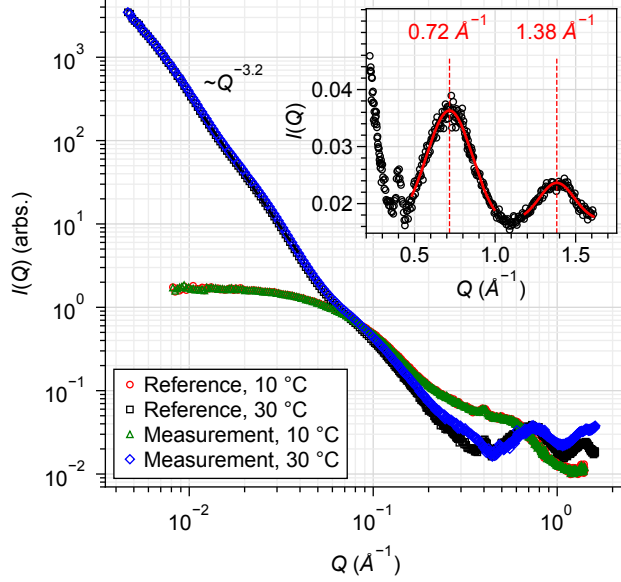


Figure 2: Temperature-dependent structure and reversibility of (VPAVG)₃₀. At 10 °C, the Measurement and Reference profiles overlap and the Reference data give an apparent $R_g = 22.65 \pm 0.50 \text{ \AA}$ (Figure S7). Heating produces an approximate $q^{-3.2}$ low- q response and peaks at $q = 1.38$ and $0.72 \text{ \AA}^{-1}$, corresponding to real-space spacings of approximately 4.5 and 8.8 $\text{\AA}$, respectively. The dashed line is an unweighted power-law fit to the Reference profile at 30 °C over $0.012 < q < 0.040 \text{ \AA}^{-1}$. Inset: the high- q region of that same profile (open circles, same arbitrary intensity units as the main panel); red curves are independent Gaussian-plus-linear-background fits over $0.48 < q < 1.00$ and $1.18 < q < 1.70 \text{ \AA}^{-1}$, and the red dashed lines and labels mark the fitted peak centers. The recovery and reproducible re-formation of these signatures demonstrate a thermally reversible structural transition within experimental sensitivity.

scattering signifies self-similar density heterogeneity over the probed interval rather than a single resolved structural scale.³¹ Related gel-forming ELPs exhibit disordered bicontinuous or fractal protein-rich networks produced by arrested phase separation.^{20,21} The 4.5 and 8.8 Å spacings are compatible with strand-to-strand and sheet-to-sheet packing in cross- β -like assemblies.^{6,32} Both features are broad and exhibit full widths at half maximum of 0.25 and 0.34 Å⁻¹ (correlation lengths $2\pi/\Delta q$ of only about 25 and 19 Å), so the ordering extends over a few repeat spacings rather than forming extended cross- β crystallites. Their concurrent appearance with the low- q response establishes coexistence of local ordered contacts and mesoscale density heterogeneity, but does not by itself prove that the contacts are the network junctions. Hydrated or weakly packed β -sheet interfaces can form dynamic hydrogel contacts and reversible fibrils.¹¹⁻¹³ The disappearance of the peaks on cooling therefore identifies thermally labile β -sheet-like associations that are plausible, rather than uniquely established, physical junctions.

To resolve the evolution at fixed thermal driving force, a freshly loaded sample was brought to 30 °C on the slow ramp given in the Experimental Section. The waiting time t_w was defined from the moment the calibrated RTD readout reached 30 °C, which coincides with the start of the first isothermal acquisition. The sample was then followed by simultaneous SAXS and SA-XPCS for up to 7863 s (Figure 3). Each reported t_w labels a group of 63–200 consecutive 2 s acquisitions, each at a previously unexposed position, that were averaged together; t_w is the acquisition time of the first member of the group, and a group spans 7–16 min. The SAXS intensity initially lies close to the 6 °C reference profile and subsequently rises by more than two orders of magnitude at low q , while remaining broad and monotonic (Figure 3a and Figure S8). This evolution documents increasing mesoscale density contrast without selection of a narrow domain spacing. Both the elapsed-time and the q range analyzed below are set by photon statistics rather than by a change in the sample’s dynamics. Before $t_w = 5039$ s the low- q coherent intensity is too weak for a two-step correlation function to be resolved above the noise. The temperature-ramp data of Figure S6c show the still-mobile limit directly: at

31.9°C the correlation decays completely to its baseline within the accessible delay range, so no second, slower step is present there to resolve. Above $q = 0.00827 \text{ \AA}^{-1}$ the scattered intensity has fallen by more than an order of magnitude (Figure 3a) and the corresponding g_2 is dominated by counting noise. Reliable correlation functions are therefore restricted to $t_w \geq 5039 \text{ s}$ and the five lowest q bins.

The intensity autocorrelation function is

$$g_2(q, \Delta t) = \frac{\langle I(q, t)I(q, t + \Delta t) \rangle_t}{\langle I(q, t) \rangle_t^2}, \quad (1)$$

where $I(q, t)$ is the coherent scattering intensity and Δt is the delay time. Through the Siegert relation, g_2 reports the relaxation of density fluctuations at length scale $2\pi/q$. At $q = 0.00376 \text{ \AA}^{-1}$ (approximately 167 nm), the correlation functions show a two-step decay: an initial millisecond-scale relaxation followed by an apparent plateau that persists throughout the 2 s acquisition (Figure 3b). With increasing t_w , the amplitude of the first decay decreases and the plateau rises. The fraction of fast dynamics therefore decreases while the fraction of dynamics arrested on the experimental time window increases, rather than all relaxation times shifting uniformly. Such two-step relaxation and growth of a slow component are characteristic of heterogeneous kinetic arrest in protein and colloidal gels.^{22,25,33}

The correlation functions were parameterized by

$$g_2(q, \Delta t) = 1 + \beta \left[f \exp \left[- \left(\frac{\Delta t}{\tau_{\text{fast}}} \right)^{p_{\text{fast}}} \right] + (1 - f) \exp \left[- \left(\frac{\Delta t}{\tau_{\text{slow}}} \right)^{p_{\text{slow}}} \right] \right]^2, \quad (2)$$

where β is the instrumental speckle contrast, measured independently on a static reference and held fixed (Figure S4), f is the fitted fraction of fast dynamics, $1 - f$ is the fitted fraction of slow dynamics, τ_{fast} and τ_{slow} are effective relaxation times, and p_{fast} and p_{slow} are stretching exponents. At each elapsed time, the five measured q bins were fitted globally with the stretching exponents shared across q . This phenomenological decomposition describes the observed first decay and persistent plateau without assigning either mode to a unique molecular

mechanism. Because τ_{slow} frequently exceeds the acquisition window, it is an extrapolated descriptor rather than a directly sampled terminal time; the primary quantitative descriptor is the directly constrained fast fraction f .

The fitted fast fraction decreases monotonically at every measured q , from $f = 0.42$ – 0.71 at $t_w = 5039$ s to $f = 0.00$ – 0.13 at $t_w = 7863$ s (Figure 3c). Equivalently, the fitted slow fraction $1 - f$ rises from 0.29 – 0.58 to 0.87 – 1.00 over approximately 47 min. The common trend across $q = 0.00376$ – 0.00827 Å⁻¹ (approximately 76–167 nm) shows that the loss of mobility spans the measured mesoscale range: an increasing fraction of the scattering ensemble is dynamically arrested within the accessible delay range. Moreover, at every analyzed time f increases monotonically and significantly with q (adjacent- q differences of 1.8 – 2.9σ at $t_w = 5039$ s and 8σ or more thereafter; across the full q range the increase is 9σ or more at every analyzed time), so arrest develops first and most strongly at the largest length scale probed (167 nm) and progressively less so toward 76 nm. This length-scale ordering links the growing low- q SAXS structure directly to the arrest fraction within the same acquisition, independent of elapsed time, and is consistent with coarsening in which the largest structural features become the least mobile. XPCS does not locate those dynamically arrested regions, and $1 - f$ is not a direct geometric volume fraction because the scattering contribution is weighted by structure and contrast. The stretching exponents remain below unity. $\tau_{\text{fast}}(q)$ decreases with q with an approximately diffusive-like exponent at every analyzed time, $\gamma_{\text{fast}} = -2.0$ to -2.4 , although at the final time the fast mode is no longer detected at the lowest q and the exponent is correspondingly less precise; τ_{slow} is not used for mechanistic scaling because it lies beyond the acquisition window (Figure S9). Correlation functions at the remaining q values are shown in Figure S10. Fresh-position acquisition and an independent higher-flux control, the latter on a more dilute and still-mobile sample, show no systematic flux-dependent change in the correlation decay (Figure S5 and SI Section 5).

The simultaneous growth of low- q SAXS intensity and decline of the fast fraction establish complementary aspects of network formation. This coupling is consistent with network-

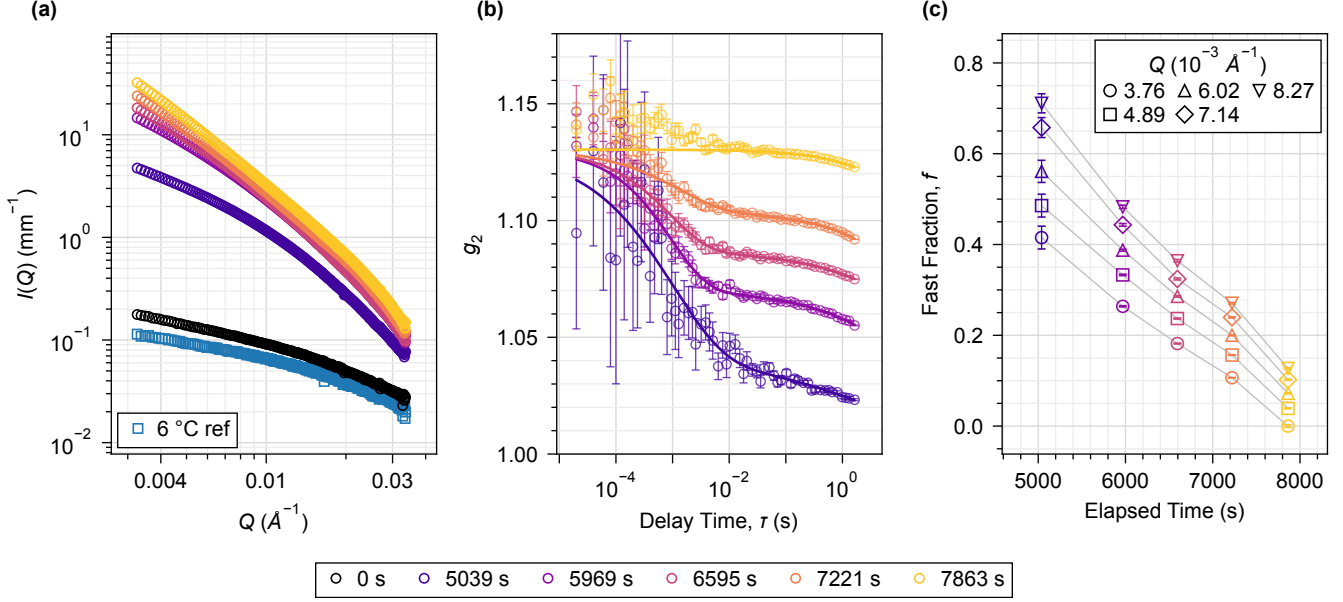


Figure 3: Isothermal structure evolution and dynamic arrest of (VPAVG)₃₀ at 30 °C. (a) Buffer-subtracted SAXS on an absolute scale (absolute scattering cross section per unit volume, calibrated as in Figure S3), showing continued growth of low- q density contrast. Panel (a) alone also carries the 6 °C reference (blue squares, keyed separately) and the 0 s profile; (b) and (c) cover the five later times only. Full series in Figure S8. (b) Representative $g_2(q, \Delta t)$ at $q = 0.00376$ Å⁻¹; solid lines are global two-mode fits, and the error bars are the measured pointwise g_2 uncertainties. With elapsed time, the amplitude of the first decay decreases and the persistent plateau rises. Correlation functions and fits at the four other q values are given in Figure S10. (c) The fitted fraction of fast dynamics, f , decreases at all five q values; equivalently, the fitted slow fraction, $1 - f$, increases. Grey lines connect points at the same q and are guides to the eye. Error bars in (c) are one-standard-deviation fit uncertainties. The underlying stretching exponents and relaxation-time scaling are given in Figure S9. Colors denote elapsed time throughout, keyed along the bottom; each elapsed time labels a group of averaged acquisitions as defined in the text.

forming arrested phase separation in a dynamically asymmetric protein–solvent mixture, for which concentration fluctuations produce a connected dense phase and viscoelastic constraints suppress relaxation.^{24,25,27,34} Related ELPs form bicontinuous networks with long stress-relaxation times and gel-like elasticity.^{20,21,35} The present measurements add a local structural element from a separate experiment: the high-temperature state of (VPAVG)₃₀ contains thermally reversible β -sheet-like contacts, and, in the isothermal experiment, the low- q assembly progressively loses mobility.

Four limitations define the appropriate interpretation. First, the β -sheet-like assignment is based on WAXS peak positions and is not an independent spectroscopic determination of secondary structure. Second, the high- q ordering and low- q XPCS dynamics were measured in separate experiments, so the data do not establish a time-resolved causal relation between peak growth and arrest. Third, the accessible delay range of the 2 s acquisition, 2×10^{-5} –1.6 s, quantifies the fraction of dynamics arrested on that time scale but cannot determine the infinite-time nonergodicity factor or establish terminal arrest at arbitrarily long delay times. Fourth, the isothermal series was acquired from a single loaded sample; the fresh positions provide spatial sampling within one preparation, and the quoted uncertainties are fit uncertainties rather than between-sample variability, so the reported trend in f is a single-preparation observation. Subject to those constraints, the combined data support network-forming arrested phase separation containing candidate, thermally labile β -sheet-like junctions.

In summary, SAXS/WAXS resolves reversible reorganization of (VPAVG)₃₀ from a dispersed state to a hierarchical, network-like assembly carrying β -sheet-like local order, while, in a separate isothermal experiment, SA-XPCS shows a near-complete shift from fast to slow dynamics across 76–167 nm. The result establishes a sequence-defined protein-polymer system in which amyloid-like local packing is thermally reversible even as the mesoscale assembly becomes dynamically arrested at high temperature.

Experimental Section. The Sanger-verified expression construct encoded an N-terminal

His-tagged (VPAVG)₃₀ protein; the complete amino-acid sequence is given in the Supporting Information. Recombinant (VPAVG)₃₀ was expressed in *E. coli*, purified by immobilized metal-affinity chromatography, and buffer exchanged into 20 mmol L⁻¹ HEPES, 250 mmol L⁻¹ NaCl, and 2 mmol L⁻¹ DTT at pH 8. Expression, solubility, and purity were monitored by SDS-PAGE (Figure S1b). Protein concentration was determined from A_{280} using a NanoDrop spectrophotometer and the sequence-specific molar extinction coefficient used during sample preparation ($\epsilon_{280} = 1490 \text{ M}^{-1} \text{ cm}^{-1}$ for the 15,461 Da construct); the final concentration was approximately 100 mg mL⁻¹. Detailed synthesis, purification, sample-cell, calibration, and fitting procedures are provided in the Supporting Information.

Temperature-dependent SAXS/WAXS measurements were performed at APS 12-ID-B with a 13.3 keV beam of approximately $100 \times 150 \text{ } \mu\text{m}^2$ and an incident flux of approximately $5 \times 10^{11} \text{ photons s}^{-1}$. Simultaneous SAXS/WAXS exposures were 0.1 s. At each set point the capillary was held at the target temperature for at least 10 min before exposure. SAXS and WAXS were acquired using an Eiger2 9M detector at an approximately 2 m sample-to-detector distance and a Pilatus 300K detector at 0.42 m, respectively. Buffer and capillary backgrounds were subtracted, and the SAXS and WAXS profiles were stitched across their overlap. SA-XPCS was performed at APS 8-ID-I with a 10.91 keV, $10 \times 10 \text{ } \mu\text{m}^2$ partially coherent beam and a Rigaku XSPA-500k detector 7.8 m downstream. The aluminum cell was conductively coupled to an anodized-aluminum holder on a Peltier-controlled stage with a resistive heater, and a calibrated RTD adjacent to the Peltier element supplied the controller readout. Before the isothermal sequence, the sample was held at 6 °C for more than 30 min, ramped to 20 °C at 5 °C min⁻¹, and then to 30 °C at 1 °C min⁻¹ to limit overshoot; $t_w = 0$ was defined when the RTD readout reached 30 °C. Each acquisition contained 100,000 frames at 50 kHz (20 μs per frame, 2 s total). The isothermal sequence was acquired from one loaded sample; the fresh positions provide spatial sampling within that preparation rather than independent sample replication. Further acquisition, fitting, and radiation-control details are provided in the Supporting Information.

Supporting Information

Recombinant synthesis and purification; gene-construction sequences and SDS-PAGE analysis; sample environments; SAXS/WAXS calibration; SA-XPCS analysis and global fitting; speckle-contrast calibration; radiation controls; Guinier analysis; complete isothermal SAXS evolution; length-scale-dependent XPCS parameters; correlation functions at all measured q values; thermal-cycling repeatability; a still frame and link for the video of the reversible thermal transition; code and reduced-data availability; and AI-assisted development with public availability of the analysis code and data (PDF).

Data and Code Availability

Reduced data and analysis/figure-generation code are publicly available at <https://github.com/qzhang234/PubCode-2021-IDPP>. The repository includes the 8-ID-I and 12-ID-B datasets used for the figures, radiation-control data, and analysis scripts, and every data-derived figure can be regenerated from it alone. Raw beamline data are available from the corresponding authors upon reasonable request.

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AI-assisted manuscript and code preparation. The authors used Claude Sonnet

5 and Opus 5 to assist with the manuscript preparation and with development of the data-analysis and figure-generation code. The Claude models were accessed through Argo, Argonne National Laboratory’s internal generative AI interface (see Supporting Information). That code and the reduced data it operates on are publicly available at <https://github.com/qzhang234/PubCode-2021-IDPP>, so the reported results can be regenerated and checked independently. The authors take full responsibility for all manuscript content and reported results.

Conflict of Interest

The authors declare no competing financial interest.

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